Higher National Diploma in Engineering (Civil)

4th Year, First Semester Examination - 2020

CE3221 – Highway and Railway Engineering

Instructions for Candidates:
Answer five questions only

No. of questions: 06
No. of pages: 04
Time: 03 hours

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1).

a) What do you mean by highway geometric design? (4 marks)

b) Write four objectives of highway geometric design. (4 marks)

c) What are the factors (design) to be considered in highway geometric design? (4 marks)

d) A horizontal curve is to be designed to connect two straight sections of a highway. The deflection angle is 30° and the radius of the curve is fixed at 403.15 m, calculate the tangent length and long chord length. (4 marks)

e) Explain the basic elements of transportation planning. (4 marks)

(Total 20 Marks)

2).

a) Explain the factors to be considered when designing vertical curves. (4 marks)

b) "Map study" is the first stage in highway alignment and detailing. Explain four features to be included under Map Study. (4 marks)

c) Explain the importance of estimates in highway construction with respect to earthworks. (4 marks)

d) Define the following terms with respect to highways.

i. Shoulders

ii. Carriageway

iii. Curb

iv. Side slopes (8 marks)

(Total 20 Marks)

3).

a) What do you mean by "Classification of soils for highway use?" (2 marks)

b) A soil has a bulk density of 1.95 Mg/m^3, water content 9.2% and specific gravity 2.68. calculate the void ratio and degree of saturation of the soil. Assume the density of water as 1000 kg/m^3. (8 marks)

c) Explain the following terms related to the soils used in highway construction.

i. Moisture content

ii. Degree of saturation

iii. Shrinkage

iv. Permeability

v. Compressibility (5 marks)

d) Write the purposes of conducting following tests on bitumen and aggregates which are uses in road construction.

i. Float test

ii. Solubility test

iii. Ductility test

iv. Los Angeles Abrasion test

v. Aggregate Shape test (5 marks)

(Total 20 Marks)

4).

a) What are the functions of a flexible pavement? (4 marks)

b) Describe the importance of subgrade layer in road construction. (4 marks)

c) Write four factors to be considered in pavement designing. (4 marks)

d) Write four causes for structural failures in pavements. (4 marks)

e) Compare the advantages and disadvantages of flexible and rigid pavements (write 2 factors for each). (4 marks)

(Total 20 Marks)

5).

a) Assume you are a highway design engineer for a newly proposed highway construction project in a rural area. List the factors you should consider before designing a perfect drainage system. (4 marks)

b) Name four types of culverts that can be proposed for the abovementioned highway project. (4 marks)

c) Describe the importance of highway maintenance. (4 marks)

d) Road traffic accidents are huge disasters which usually happen all around the world and causes for many deaths. Many factors will affect it and one factor is poor driver behavior. List four factors that can be accounted to highlight the factor "poor driver behavior". (4 marks)

e) List three basic types of traffic control devices. (4 marks)

(Total 20 Marks)

6).

a) Describe the objectives of traffic engineering. (5 marks)

b) Define following terms with neat sketch considering vehicle movements in a traffic stream.

i. Car following

ii. Lane changing

iii. Queuing

iv. Merging

v. Diverging (5 marks)

c) Define the following terms used in traffic engineering.

i. Average Daily Traffic

ii. Spot speed

iii. Journey speed

iv. Model speed

v. Running speed (5 marks)

d) Name five places which can be considered as, on–street parking is prohibited. (5 marks)

(Total 20 Marks)